DYNAMIC ON-THE-FLY MODIFICATIONS OF SPARK APPLICATIONS

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SPARK SUMMIT
EUROPE 2016

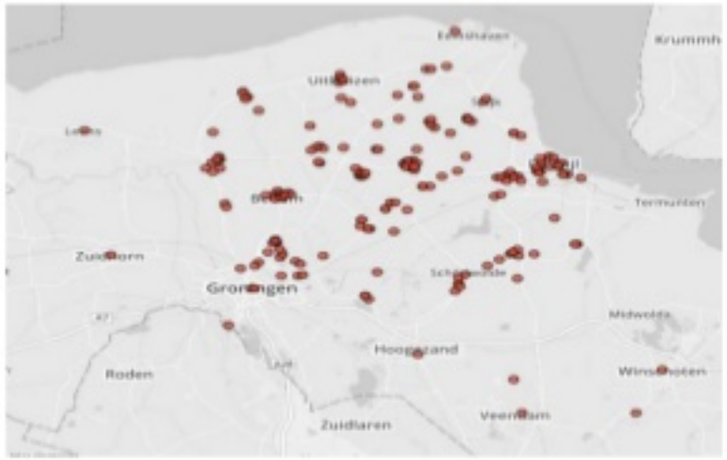
Self.me

- **Scientist innovator**
  Monitoring & Control Services group
  @TNO, from January 2012

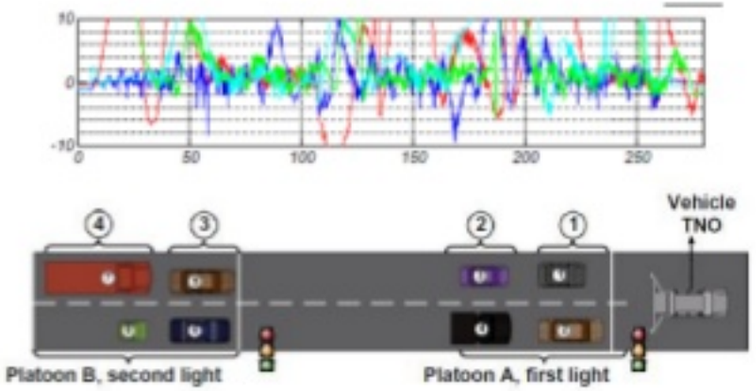
Applied Research Organization of NL

TNO innovation for life

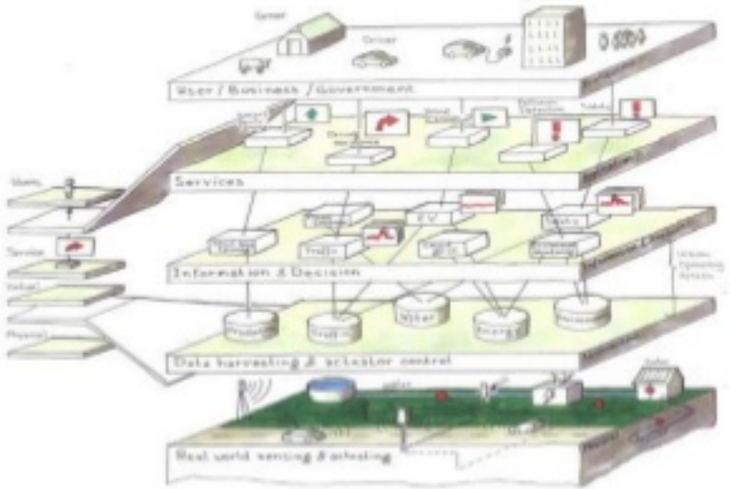
Earthquakes real-time monitoring:
350 buildings in Groningen (NL)



Cooperative automated driving



Live monitoring of water and gas pipes for whole NL



Joint work: TNO and RuG

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university of groningen

Elena Lazovik

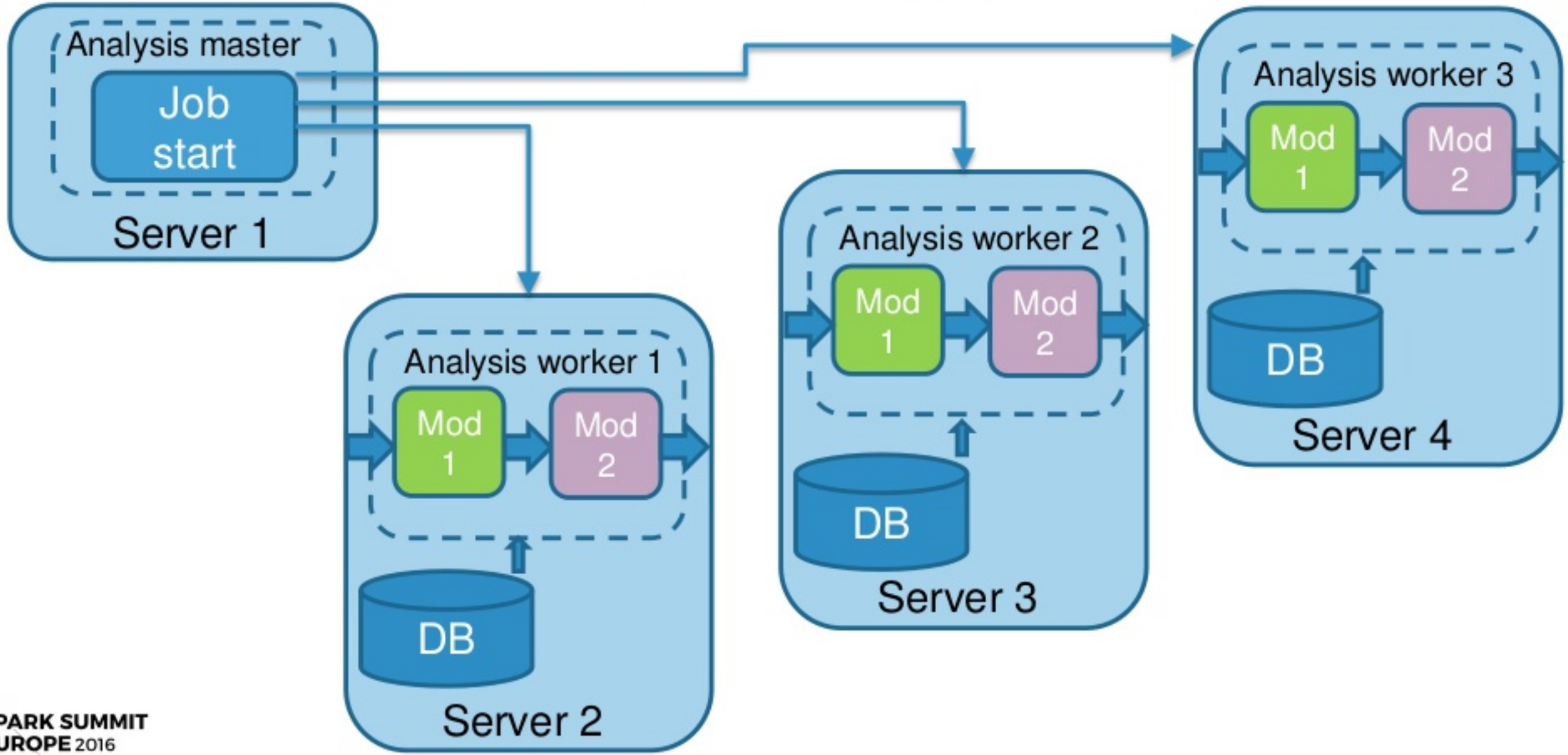
TNO innovation for life

Erik Langius

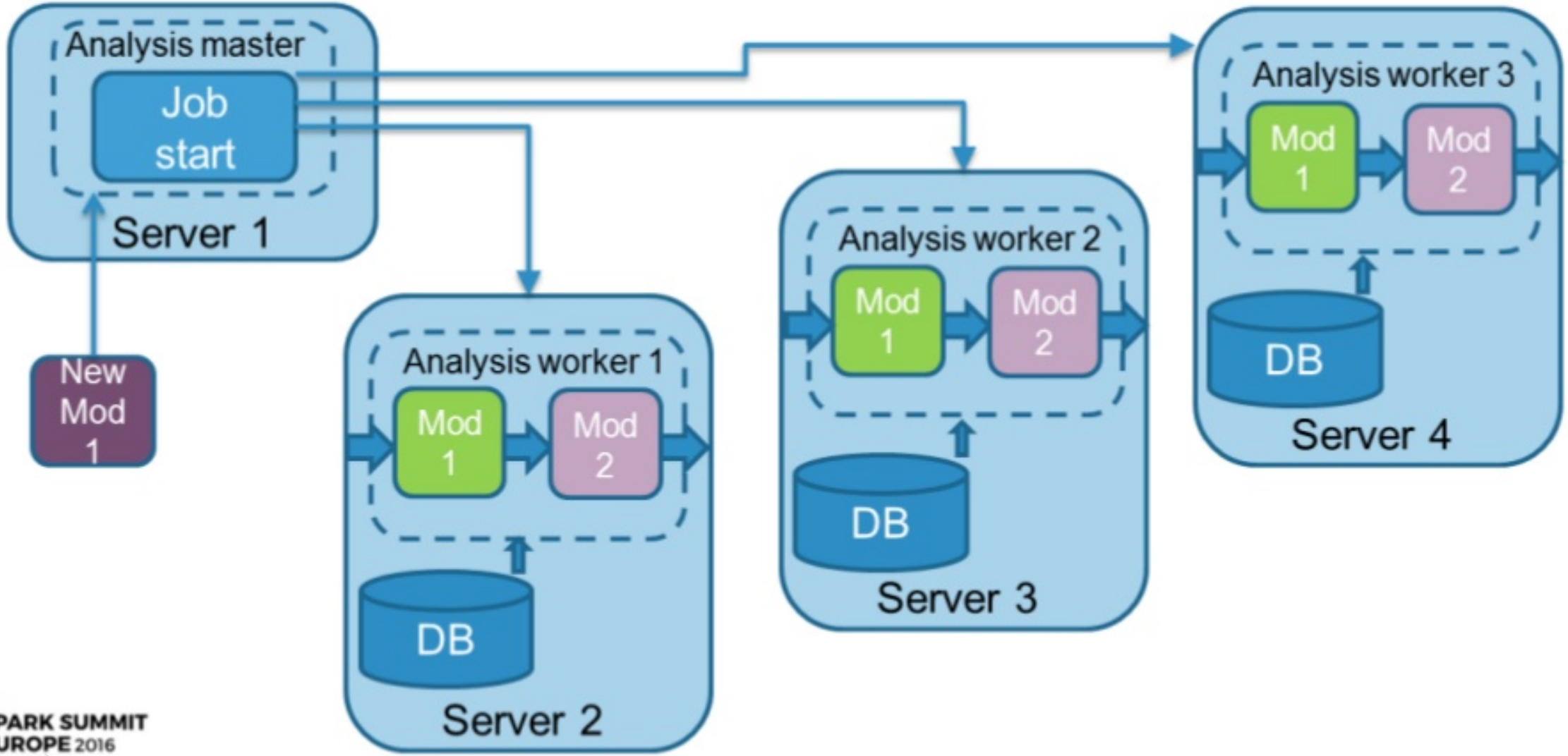
Michel Medema

Toon Albers

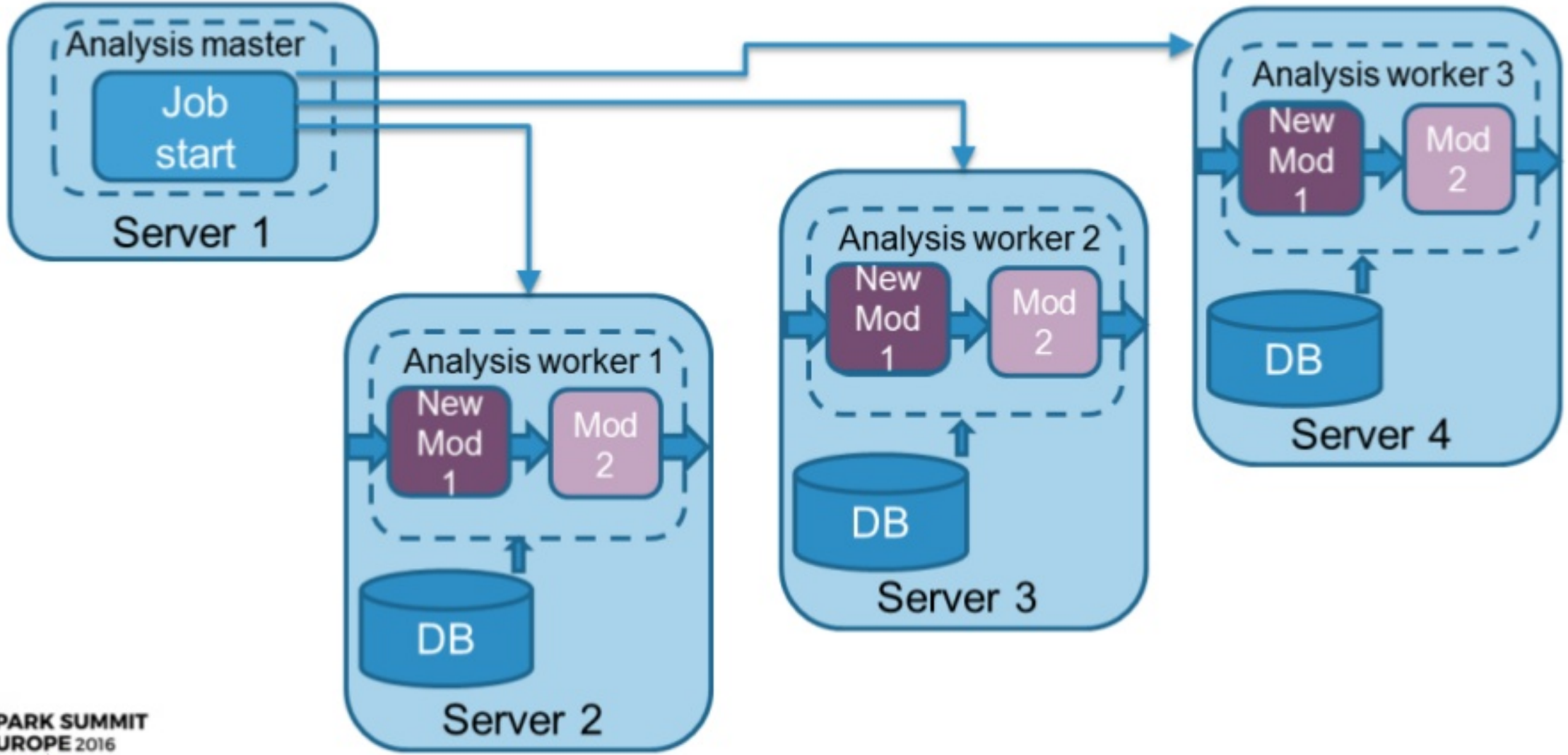
Default Spark application



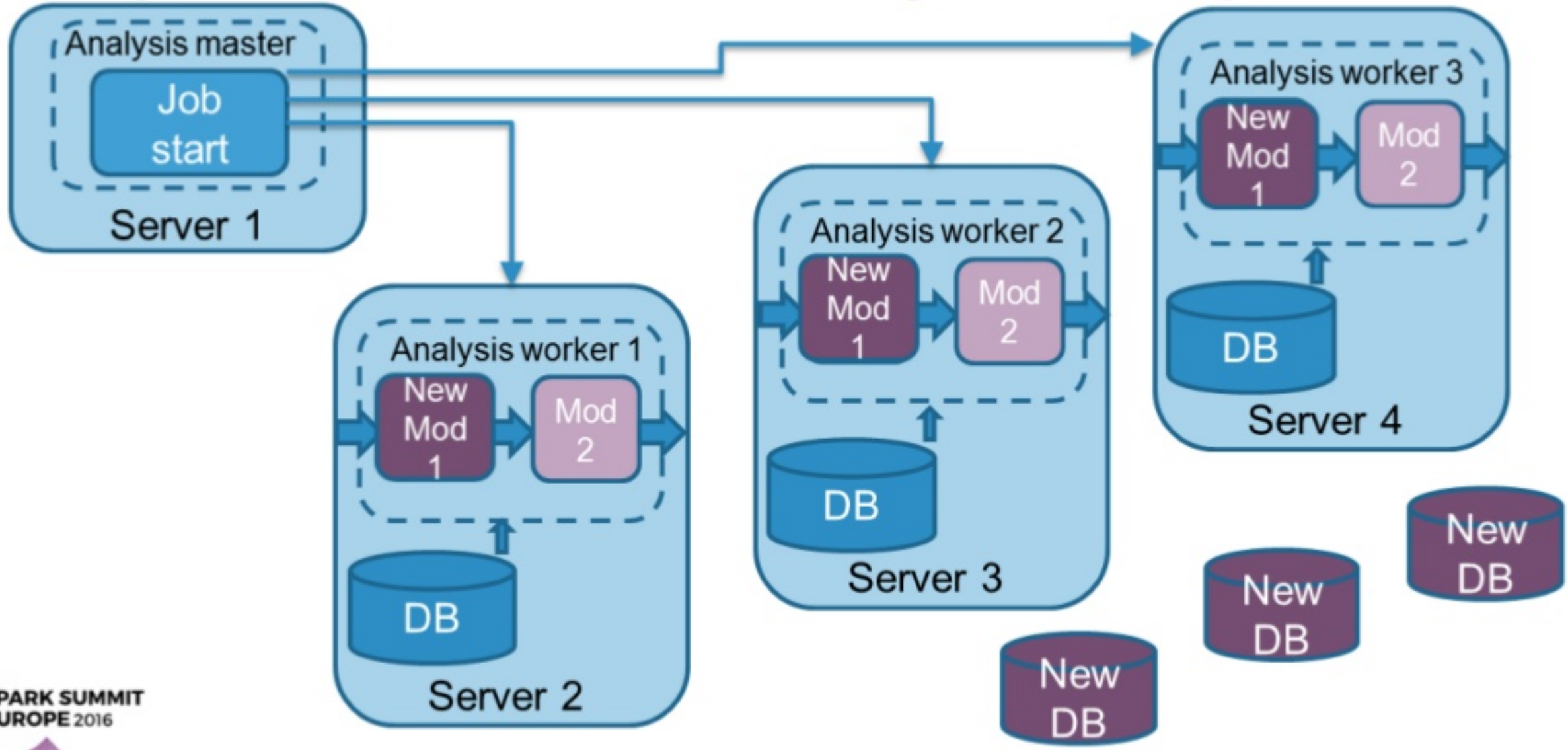
Dynamic On-the-Fly changes 1/4



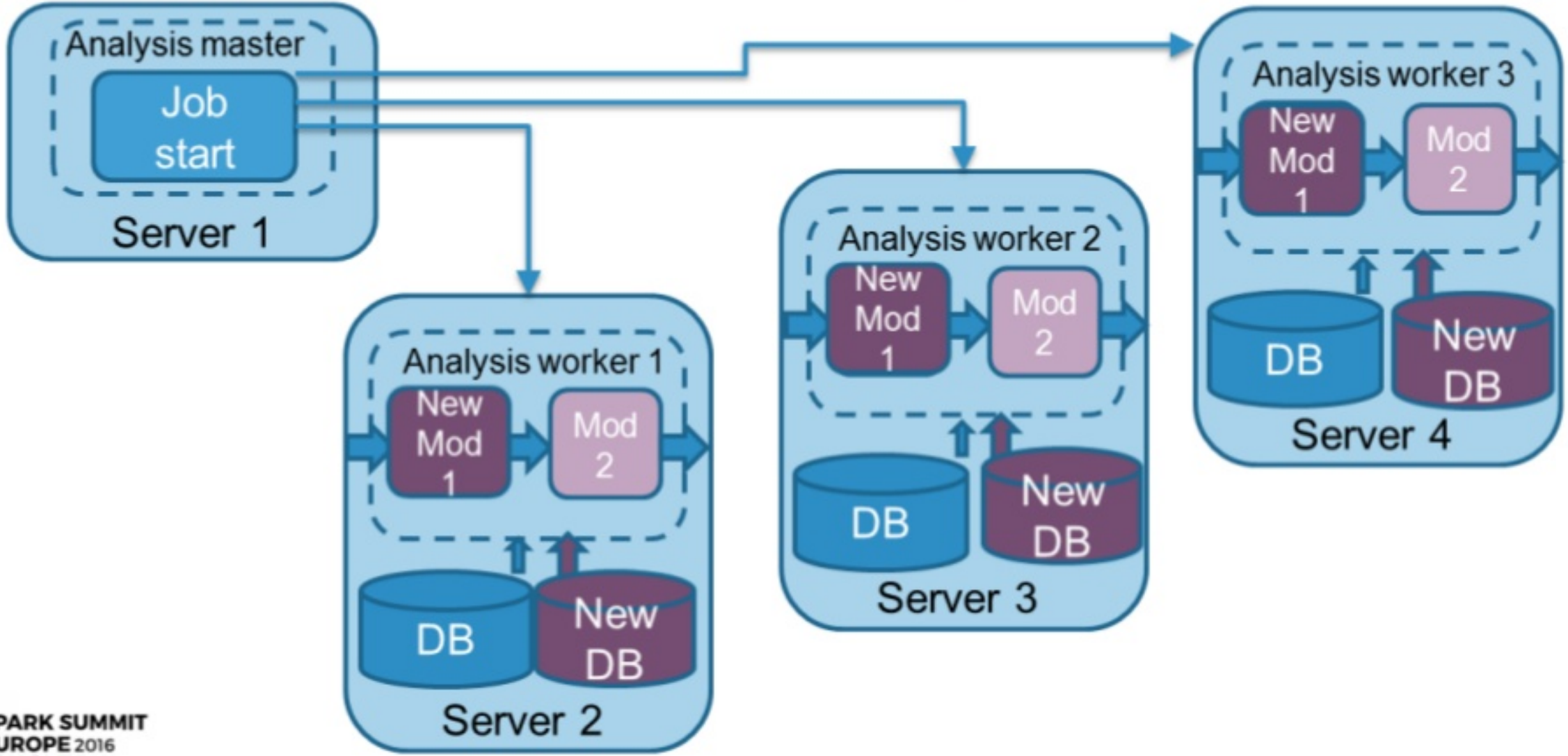
Dynamic On-the-Fly changes 2/4



Dynamic On-the-Fly changes 3/4

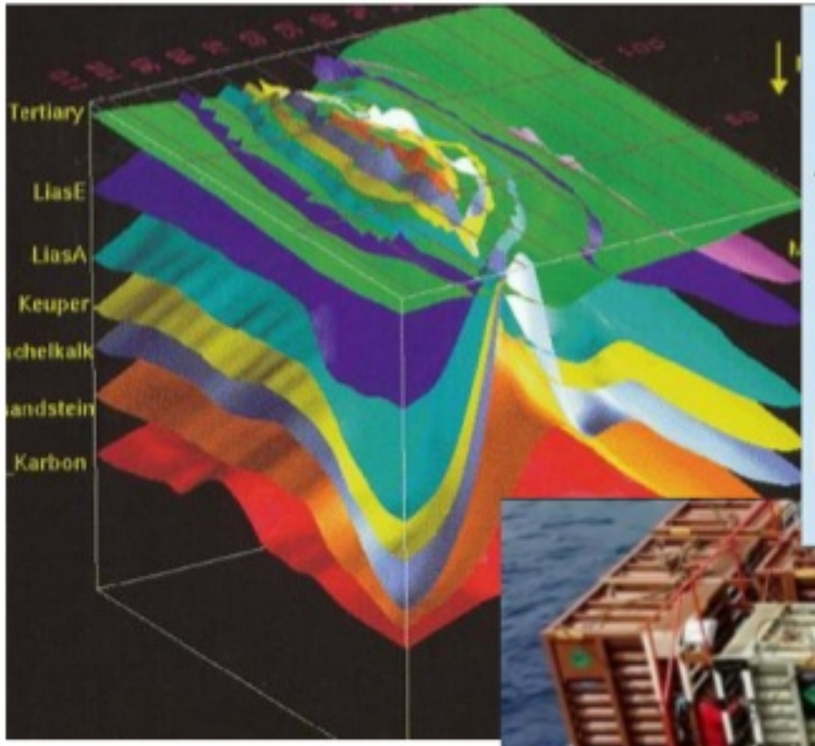


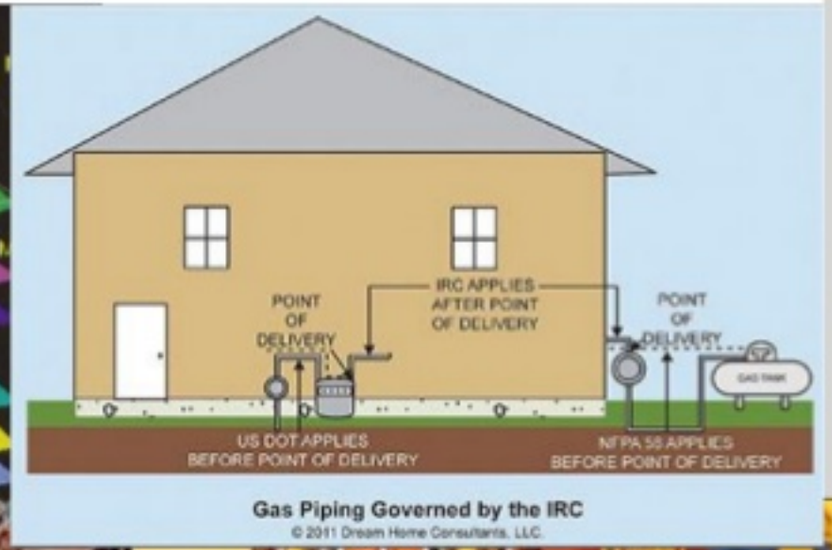
Dynamic On-the-Fly changes 4/4



Why is it needed?

Long Beach, CA





Dynamic changes of functions

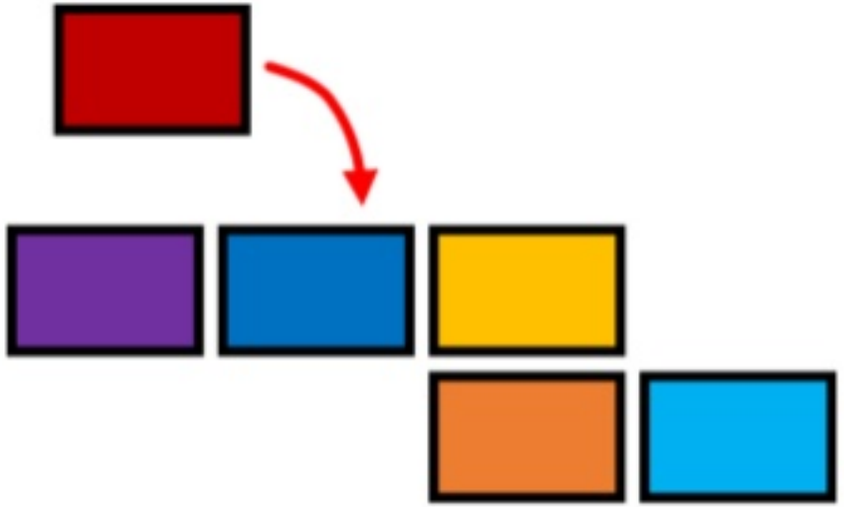
- Given a Spark driver program:
  - We want to change a function/parameter in scenario
  - We want to switch one data source to another

- without stopping the whole application!!!

What can we change?

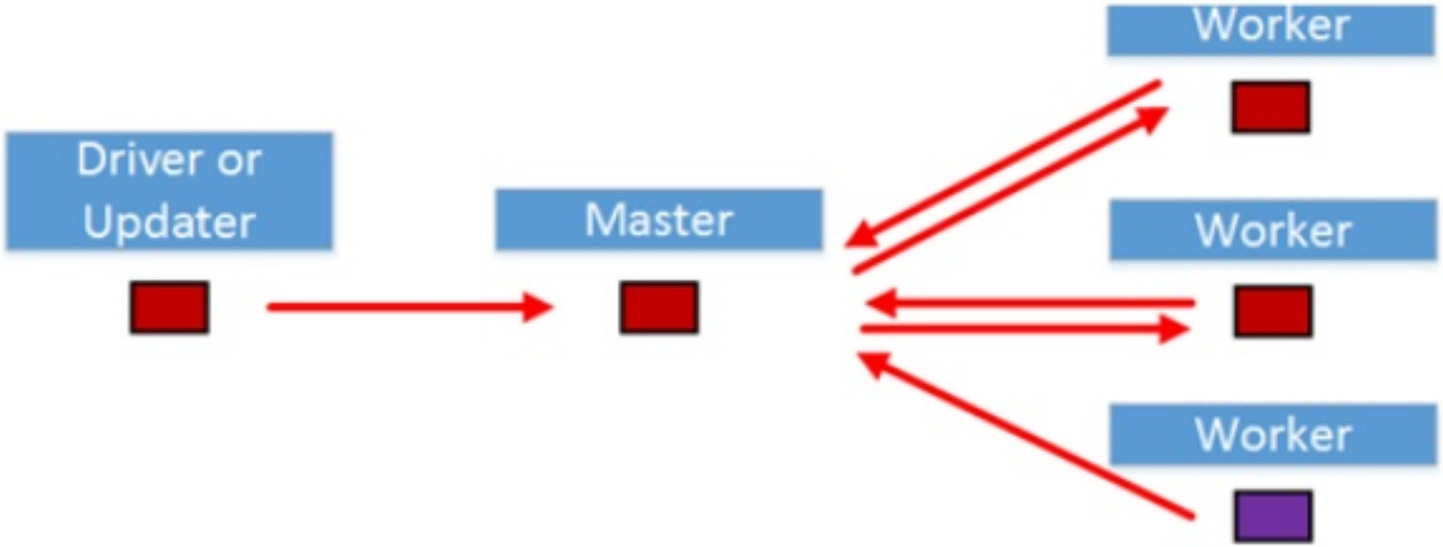
- Spark functional steps
  - `map`, `reduce`, filter, reduceByKey
    …

- Functional step content
  - Variables
  - Code (x => x + 1)



How to update?

- Push/pull?
- Variables: serialize
- Code: bytecode?



Typical calculation Spark scenario

```
val conf = new SparkConf()
val sc = new SparkContext(conf)




for (itt <- 1 until 100) {
  val rdd = sc.parallelize(1 until 1000, 10)
  val result = rdd
        .map(i => i * 10)
        .reduce(_ + _)
  println(s"$itt: $result")
}

sc.stop()
```

We want to change hardcoded parameter 10

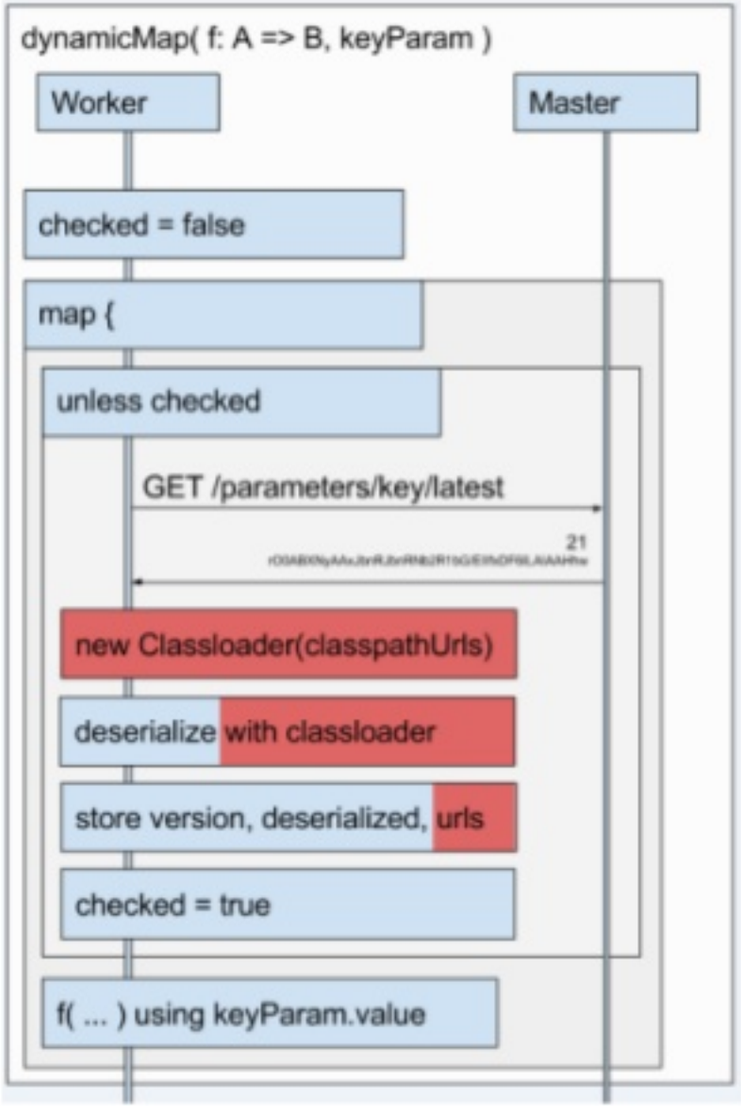
Introducing dynamic-spark

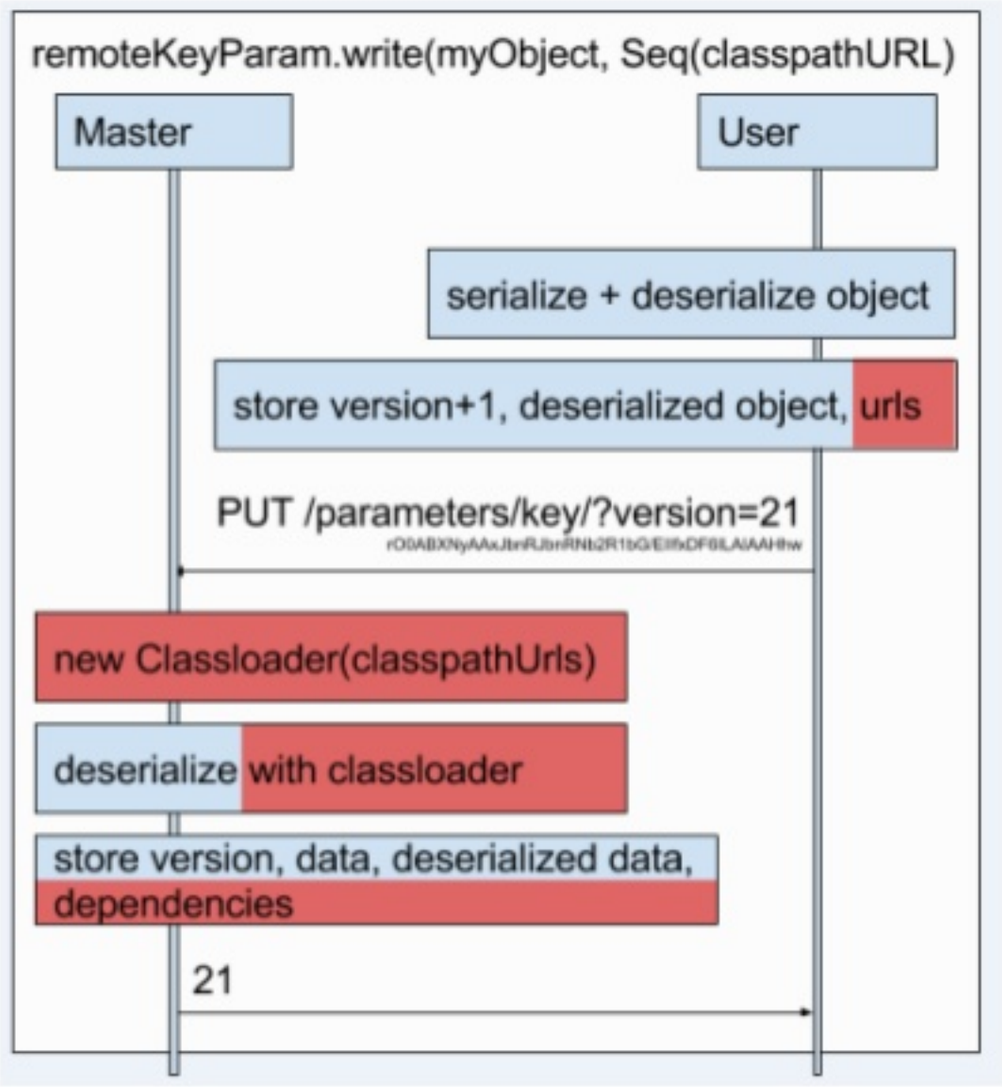
```
val conf = new SparkConf().set("dynamic.server.port", "8090")
val sc = new SparkContext(conf)
val server = new DynamicServer(conf, conf.getInt("dynamic.server.port", 0))
server.start()
val factor = new RESTParameter(server, key = "factor", version = 1, value = 10)
for (itt <- 1 until 100) {
  val rdd = sc.parallelize(1 until 1000, 10)
  val result = rdd
        .dynamicMap({ i => i * factor.value }, factor)
        .reduce(_ + _)
  println(s"$itt: $result")
}
sc.stop()
server.stop()
```

Changing parameter of the function

- val factor = new RemoteRESTParameter(serverUri, "factor", version = 1, value = 10)
- …
- factor.write(15) // set new value and increment version
- …

DynamicMap flow





Dynamic RDD functions

```
def wrap[A, V](f: A => V, params: DynamicParameter[_]*): A => V = {
  (a: A) => {
    updateParam(params)
    f(a)
  }
}

def dynamicMap(f: A => B, params: DynamicParameter[_]*): RDD[B] = {
    // Same as RDD.map, but wrapped to check for updates
    val g = wrap(f, params: _*)
    rdd.map(g)
  }
```

And dynamicReduce, dynamicFilter, dynamicReduceByKey, etc.

Passing dynamic function

```
// This object is merely defined as a place to store the functions
object FuncHolder {
  // A value referencing a function
  val compareFunc = (num:Double) => num < 0.1
  // A method that is executed by the client, which returns a
  // function that is to be serialized
  def getFunc:(Double => Boolean) = {
    (num:Double) => num < 0.2
  }
}

val funcParam = new RESTParameter[(Double) => Boolean](server, "filterfunc",
                                1, (n:Double) => n < 0 )
param.write( FuncHolder.compareFunc,    Seq("file://my/libs/func_holder.jar") )
param.write( FuncHolder.getFunc,        Seq("file://my/libs/func_holder.jar") )
// Here the bytecode is present in the user's client JAR:
param.write( (num:Double) => num < 0.3, Seq("file://this_client.jar") )
```

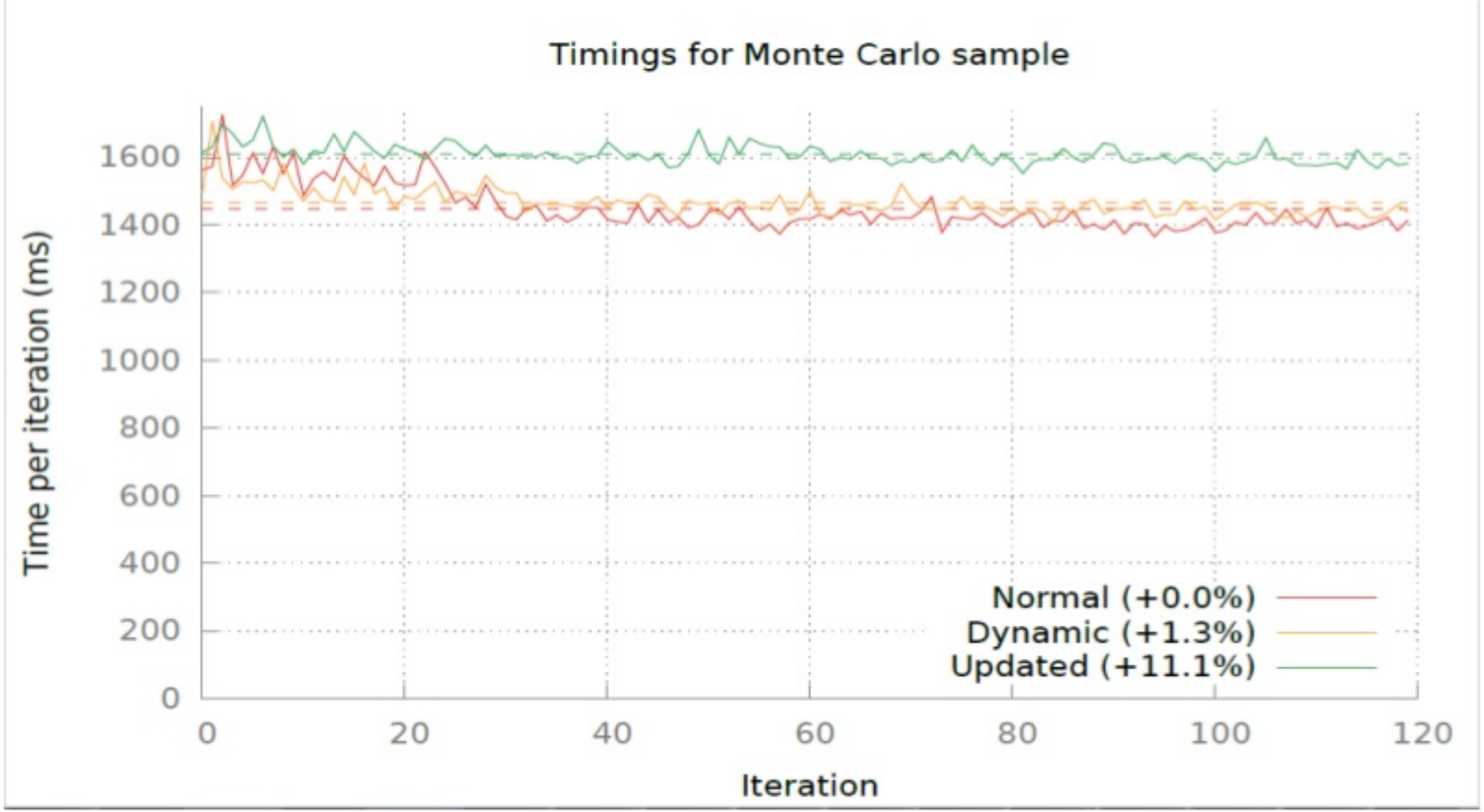
Experiments for dynamic Monte Carlo

$P_f = P(Z<0)$

Limit state function:

$$Z = R - \sum_{i=1}^{25} S_i^2 / i$$

6 physical servers



Dynamic switching of data sources

- Why use dynamic switching?
  - Data source becomes unavailable
  - More accurate data needed

- Research questions
  - Is it feasible?
  - What is the best approach?

Approaches for data sources switch

Two approaches considered

- Extending Apache Spark
- Intermediary system Solution

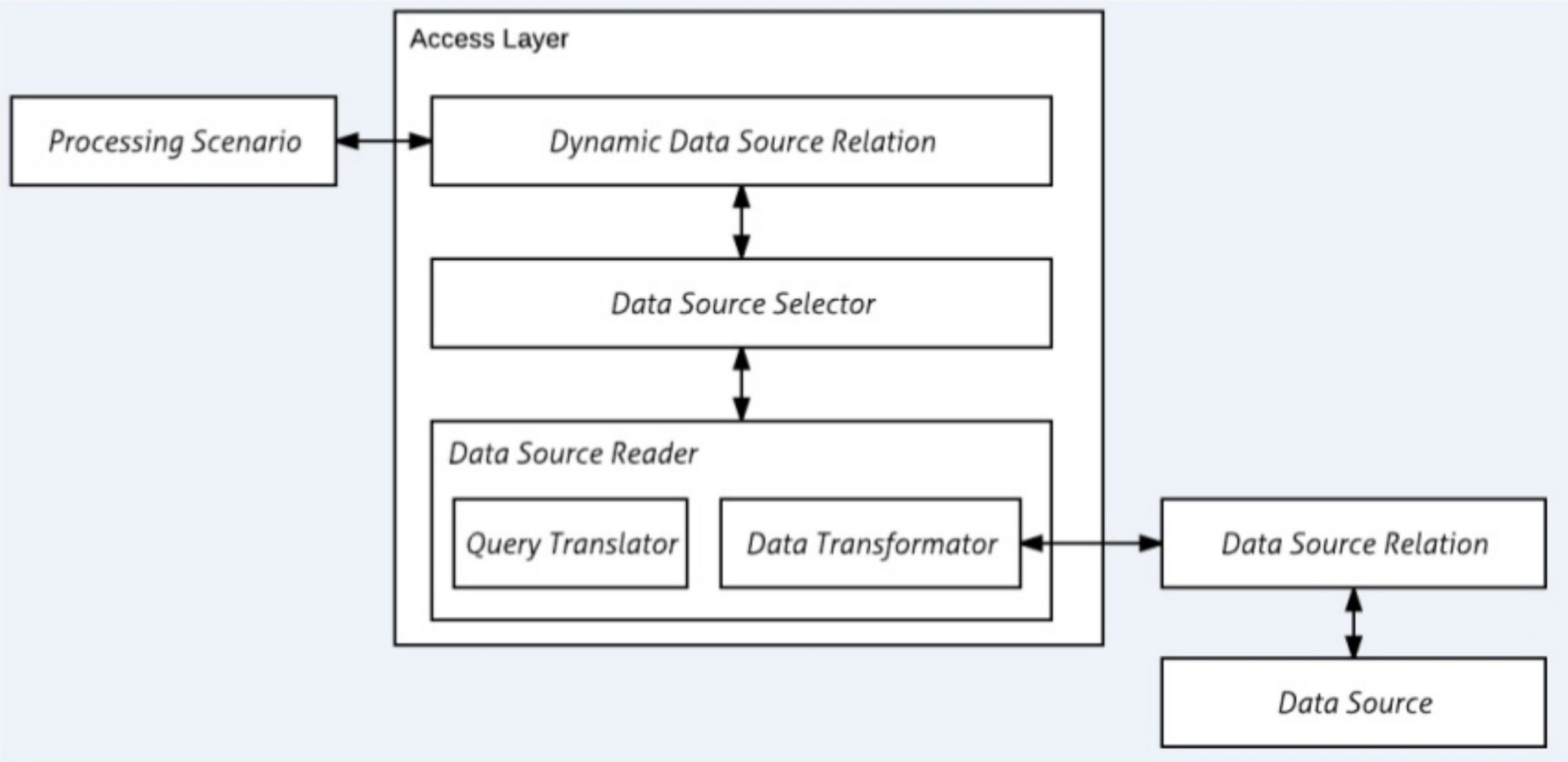
Compare based on key requirements

- Performance
- Usability
- Flexibility
- Efficiency
- Extensibility

Dynamic data sources switching

- Issues with dynamic switching
  - Heterogeneous data sources
    - Accessing Data Source
    - Query Translation
    - Data Transformation
  - Data Locality
    - Process close to data
  - Deployment

Dynamic Data Source API



Custom Spark Data Source

- Makes use of the Spark Data Source API: RelationProvider and BaseRelation classes

- Requires just two classes
  - Data Source provider, extending the "RelationProvider"
  - Data Source relation, extending e.g. "BaseRelation"

- Re-uses existing Spark data source relations

Custom Relation Provider

```
package rugds.dynamicdatasources.dynamicrelation

import org.apache.spark.sql.SQLContext
import org.apache.spark.sql.sources.{BaseRelation, RelationProvider}

private class DefaultSource extends RelationProvider {
  override def createRelation(sqlContext: SQLContext, parameters: Map[String, String]): BaseRelation = {
    new DynamicDataSourceRelation(parameters)(sqlContext)
  }
}
```

Custom Data Source Relation

```
package rugds.dynamicdatasources.dynamicrelation

import org.apache.spark.rdd.RDD
import org.apache.spark.sql.{Row, SQLContext}
import org.apache.spark.sql.sources.{BaseRelation, Filter, PrunedFilteredScan}
import org.apache.spark.sql.types._
import rugds.Logging
import rugds.dynamicdatasources.dynamicrelation.datasources.readers.DataSourceReader

case class DynamicDataSourceRelation(parameters: Map[String, String])(@transient val sqlContext: SQLContext) extends BaseRelation with PrunedFilteredScan with Logging {
  // Schema that the data should adhere to.
  val schema : StructType = StructType(Seq(
    StructField("value", DoubleType, nullable = false),
    StructField("timestamp", LongType, nullable = false)
  ))
  override def buildScan(requiredColumns: Array[String], filters: Array[Filter]): RDD[Row] = {
    // Select a data source.
    val reader: DataSourceReader = new DataSourceSelector().selectDataSource(sqlContext, schema)

    // Retrieve the RDD
    reader.read
  }
}
```

Using The Custom Source

Only **one** line different from normal usage

```
val conf = new SparkConf().setAppName("Dynamic Data Source Client")
val sc = new SparkContext(conf)
val sqlContext = new SQLContext(sc)
val rdd: RDD[String] = sqlContext.sparkContext.textFile("file.csv")
```

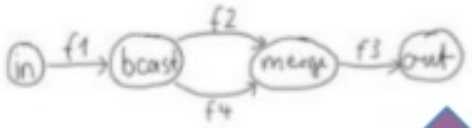
```
val conf = new SparkConf().setAppName("Dynamic Data Source Client")
val sc = new SparkContext(conf)
val sqlContext = new SQLContext(sc)
val rows: DataFrame = sqlContext.read.dynamicSource("")
```

Intermediary system version

Spark

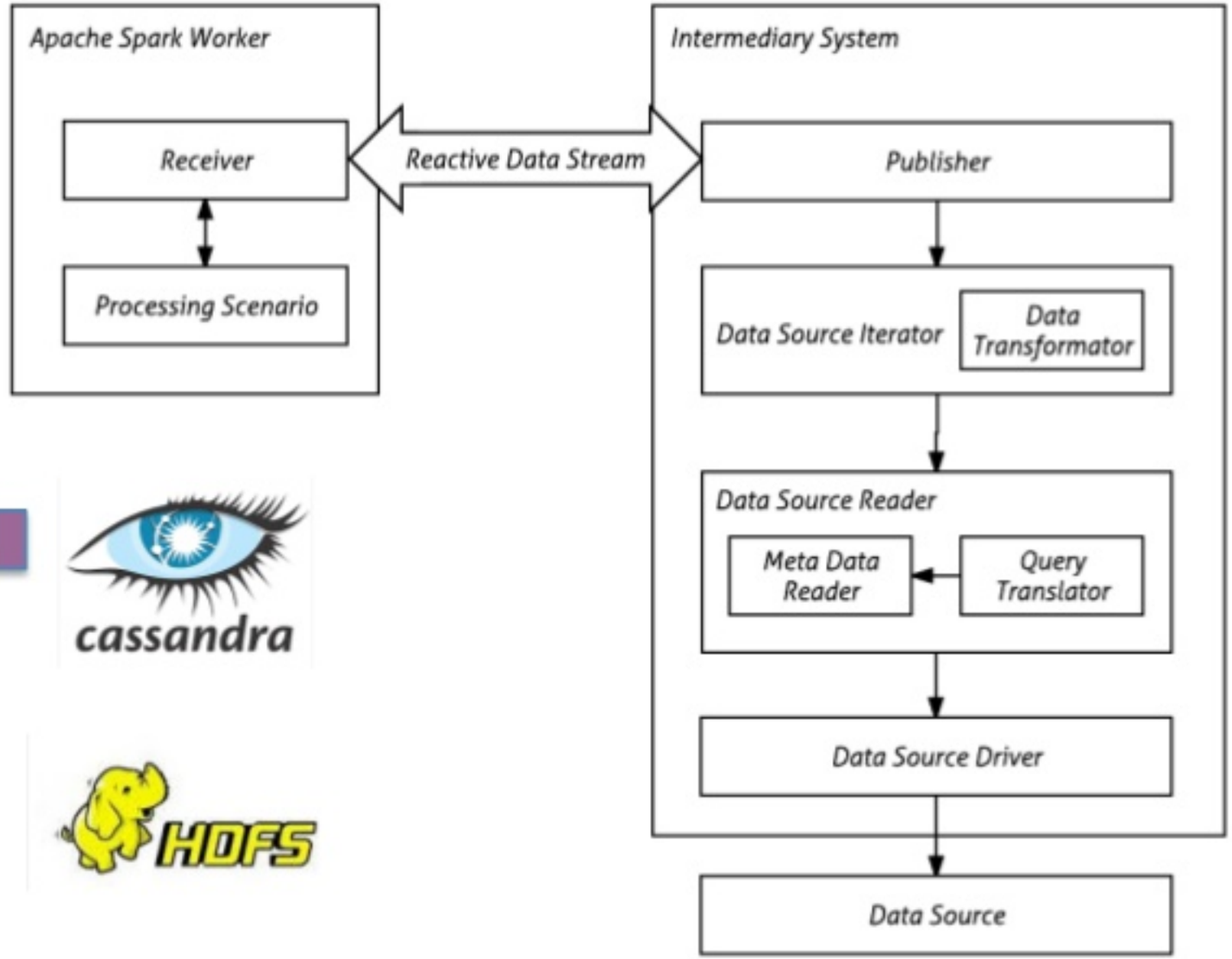
reactive stream processing with

akka streams



cassandra

HDFS



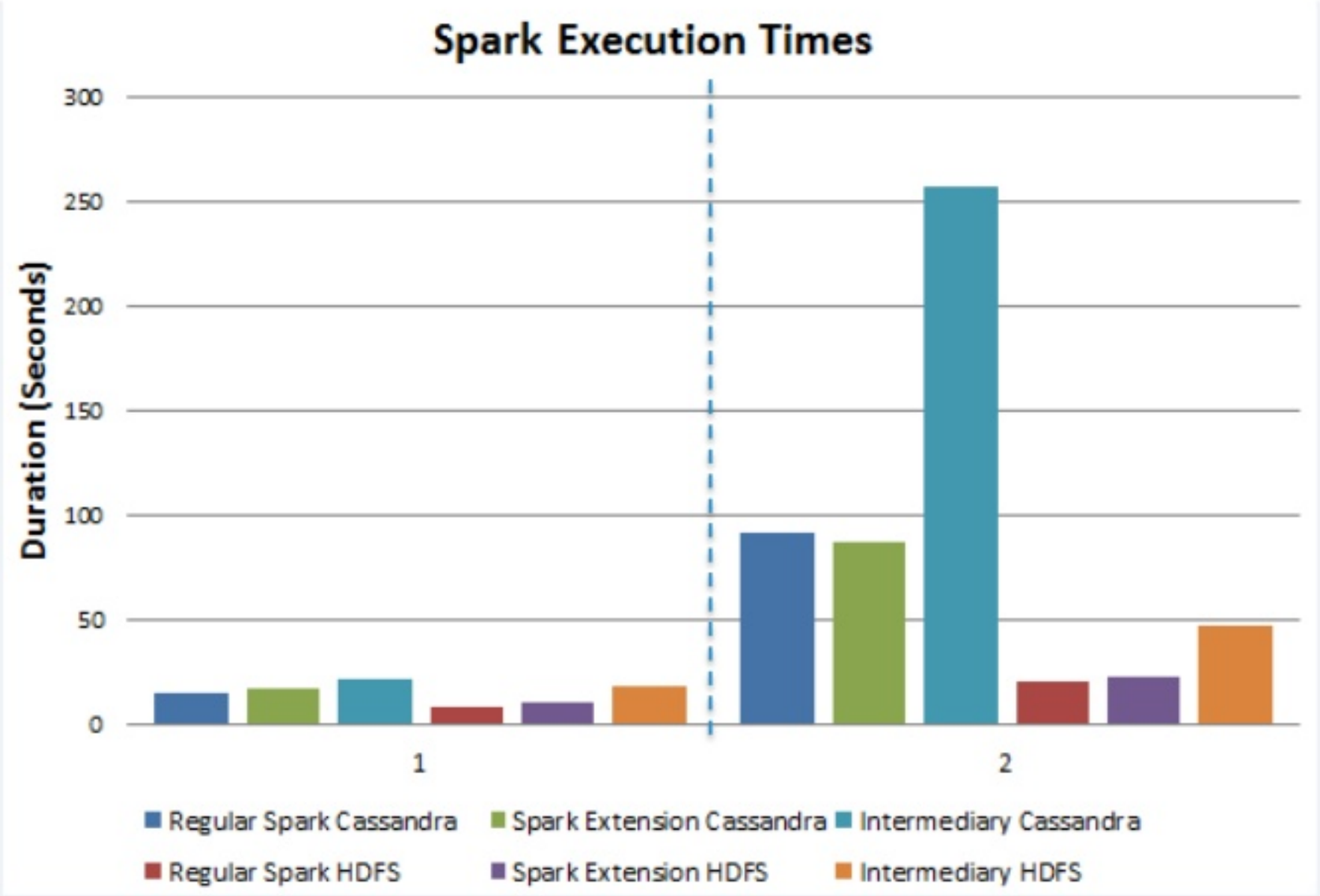
Experiments results for data sources

Data sources:
- HDFS
- Apache Cassandra

Intermediary:
reactive-streams (akka)

Two data sets
Small data set:
(58508 records)
Larger data set:
(33 million records)

6 Spark workers



Future plans

- dynamic-spark open-source library for Spark 2.0.x
  - cleaner API
  - use of datasets instead of RDD
- Data switch for streams
  - more experiments with streaming data
- More efficient dynamic code update
  - more alternatives to REST

THANK YOU.

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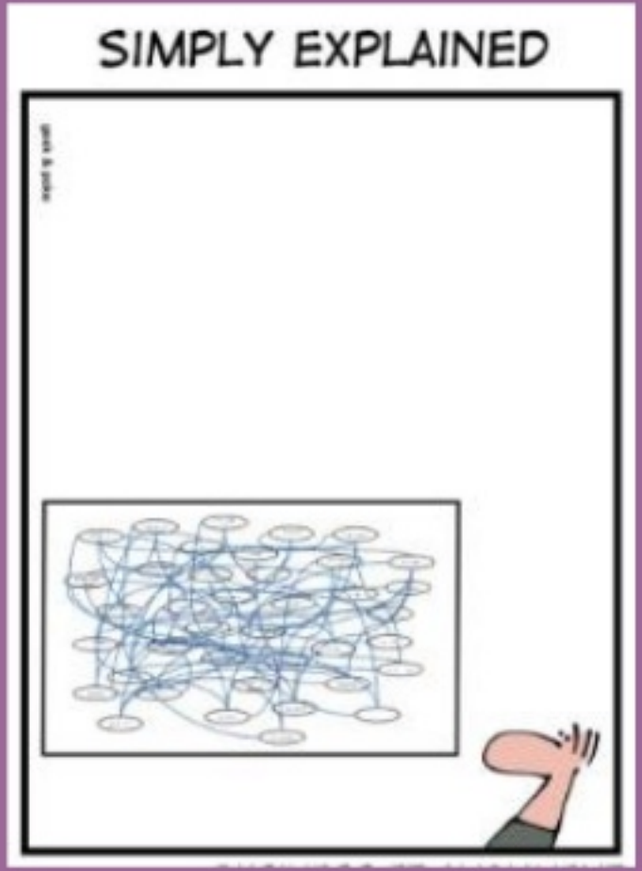

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